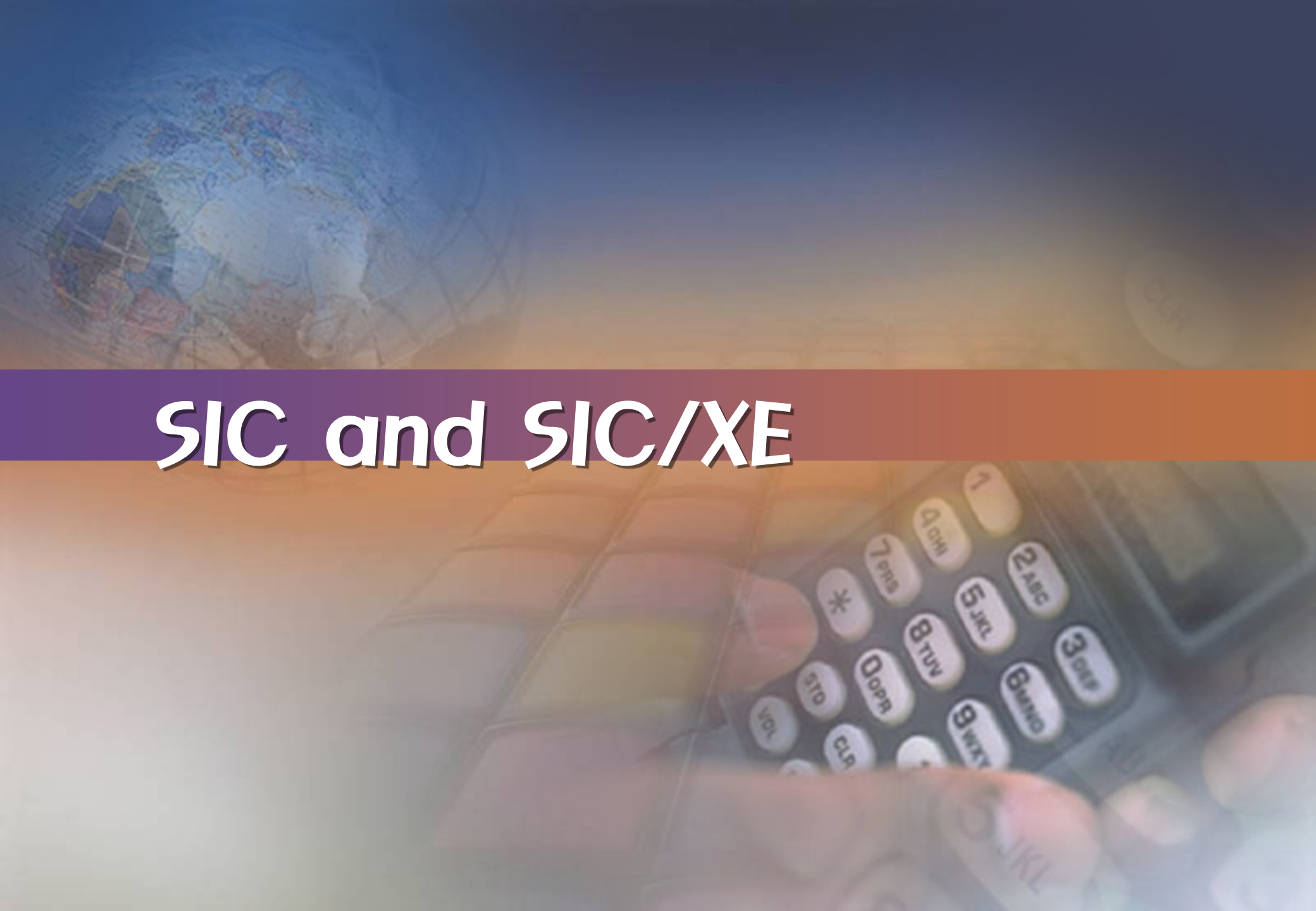


The background of the slide is a composite image. The upper left portion features a stylized, textured globe in shades of blue and white. The lower right portion shows a close-up of a hand operating a calculator, with various buttons like '1', '2', '3', '4', '5', '6', '7', '8', '9', '0', and function keys like '*' and '/' visible. A horizontal band of purple and orange gradient separates the globe from the calculator.

SIC and SIC/XE

수업 목표

- SIC/XE(Simplified Instructional Computer/Extra Expensive)의 개념에 대한 이해를 바탕으로 실제 관련 예제를 통해 이해도를 높인다.
- 다양한 SIC/XE 명령어 사용에 대해 실제 예제를 통해 이해도를 확인한다.

SIC Program Example (I-a)

- Data move operations

(ALPHA=5, C1='Z')

	LDA	FIVE	LOAD CONSTANT 5 INTO REGISTER A
	STA	ALPHA	STORE IN ALPHA
	LDCH	CHARZ	LOAD CHARACTER 'Z' INTO REGISTER A
	STCH	C1	STORE IN CHARACTER VARIABLE C1
	.		
	.		
ALPHA	RESW	1	ONE-WORD VARIABLE
FIVE	WORD	5	ONE-WORD CONSTANT
CHARZ	BYTE	C'Z'	ONE-BYTE CONSTANT
C1	RESB	1	ONE-BYTE VARIBALE

Fig 1.2(a) Sample data movement operation for SIC

SIC Program Example (I-b)

(ALPHA=5, C1='Z')

*값 5와 ASCII 코드 값 90(문자 Z)는
SIC/XE의 즉시 주소 지정 방식(#)으로 처리*

	LDA	#5	LOAD VALUE 5 INTO REGISTER A
	STA	ALPHA	STORE IN ALPHA
	LDA	#90	LOAD ASCII CODE FOR 'Z' INTO REG A
	STCH	C1	STORE IN CHARACTER VARIABLE C1
	.		
	.		
	.		
ALPHA	RESW	1	ONE-WORD VARIABLE
C1	RESB	1	ONE-BYTE VARIBALE

Using immediate addressing

Fig 1.2(b) Sample data movement operation for SIC/XE

SIC Program Example (II-a)

- Arithmetic operations

(BETA=ALPHA+INCR-1, DELTA=GAMMA+INCR-1)

	LDA	ALPHA	LOAD ALPHA INTO REGISTER A
	ADD	INCR	ADD THE VALUE OF INCR
	SUB	ONE	SUBTRACT 1
	STA	BETA	STORE IN BETA
	LDA	GAMMA	LOAD GAMMA INTO REGISTER A
	ADD	INCR	ADD THE VALUE OF INCR
	SUB	ONE	SUBTRACT 1
	STA	DELTA	STORE IN DELTA
	.		
ONE	WORD	1	ONE-WORD CONSTANT
.			ONE-WORD VARIABLES
ALPHA	RESW	1	
BETA	RESW	1	
GAMMA	RESW	1	
DELTA	RESW	1	
INCR	RESW	1	

Fig 1.3(a) Sample arithmetic operations for SIC

SIC Program Example (II-b)

(BETA = ALPHA + INCR - 1, DELTA = GAMMA + INCR - 1)

LDS	INCR	LOAD VALUE OF INCR INTO REGISTER S
LDA	ALPHA	LOAD ALPHA INTO REGISTER A
ADDR	S, A	ADD THE VALUE OF INCR
SUB	#1	SUBTRACT 1
STA	BETA	STORE IN BETA
LDA	GAMMA	LOAD GAMMA INTO REGISTER A
ADDR	S, A	ADD THE VALUE OF INCR
SUB	#1	SUBTRACT 1
STA	DELTA	STORE IN DELTA

.
.

ONE WORD VARIABLES

ALPHA	RESW	1
BETA	RESW	1
GAMMA	RESW	1
DELTA	RESW	1
INCR	RESW	1

ADDR r1, r2 format(2) opcode(90)

Effect: $r2 \leftarrow (r2) + (r1)$

Notes: X (sic/xe)

Fig 1.3(b) Sample arithmetic operations for SIC/XE

SIC/XE instruction Set 1

Appendix A

SIC/XE Instruction Set and Addressing Modes

각각의 명령은
명령어, 형식, 16진코드,
의미로 구성된다.

Instruction Set

In the following descriptions, uppercase letters refer to specific registers. The notation m indicates a memory address, n indicates an integer between 1 and 16, and $r1$ and $r2$ represent register identifiers. Parentheses are used to denote the contents of a register or memory location. Thus $A \leftarrow (m..m+2)$ specifies that the contents of the memory locations m through $m+2$ are loaded into register A; $m..m+2 \leftarrow (A)$ specifies that the contents of register A are stored in the word that begins at address m .

The letters in the Notes column have the following meanings:

- P Privileged instruction
- X Instruction available only on XE version
- F Floating-point instruction
- C Condition code CC set to indicate result of operation (<, =, or >)

The Format column indicates which SIC/XE instruction format is to be used in assembling each instruction; 3/4 means that either Format 3 or Format 4 can be used. All instructions for the standard version of SIC are assembled using the format described in Section 1.3.1 (which is compatible with Format 3). Instruction subfields that are not required, such as the address field for an RSUB instruction, are set to zero.

Mnemonic	Format	Opcode	Effect	Notes
ADD m	3/4	18	$A \leftarrow (A) + (m..m+2)$	
ADDF m	3/4	58	$F \leftarrow (F) + (m..m+5)$	X F
ADDR r1,r2	2	90	$r2 \leftarrow (r2) + (r1)$	X
AND m	3/4	40	$A \leftarrow (A) \& (m..m+2)$	
CLEAR r1	2	B4	$r1 \leftarrow 0$	X
COMP m	3/4	28	$(A) : (m..m+2)$	C
COMPF m	3/4	88	$(F) : (m..m+5)$	X F C
COMPR r1,r2	2	A0	$(r1) : (r2)$	X C
DIV m	3/4	24	$A \leftarrow (A) / (m..m+2)$	
DIVF m	3/4	64	$F \leftarrow (F) / (m..m+5)$	X F
DIVR r1,r2	2	9C	$r2 \leftarrow (r2) / (r1)$	X
FIX	1	C4	$A \leftarrow (F)$ [convert to integer]	X F
FLOAT	1	C0	$F \leftarrow (A)$ [convert to floating]	X F
HIO	1	F4	Halt I/O channel number (A)	P X
J m	3/4	3C	$PC \leftarrow m$	
JEQ m	3/4	30	$PC \leftarrow m$ if CC set to =	
JGT m	3/4	34	$PC \leftarrow m$ if CC set to >	
JLT m	3/4	38	$PC \leftarrow m$ if CC set to <	
JSUB m	3/4	48	$L \leftarrow (PC); PC \leftarrow m$	
LDA m	3/4	00	$A \leftarrow (m..m+2)$	
LDB m	3/4	68	$B \leftarrow (m..m+2)$	X
LDCH m	3/4	50	A [rightmost byte] $\leftarrow (m)$	
LDF m	3/4	70	$F \leftarrow (m..m+5)$	X F
LDL m	3/4	08	$L \leftarrow (m..m+2)$	
LDS m	3/4	6C	$S \leftarrow (m..m+2)$	X
LDT m	3/4	74	$T \leftarrow (m..m+2)$	X
LDX m	3/4	04	$X \leftarrow (m..m+2)$	
LPS m	3/4	D0	Load processor status from information beginning at address m (see Section 6.2.1)	P X
MUL m	3/4	20	$A \leftarrow (A) * (m..m+2)$	

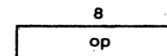
SIC/XE instruction Set 2

Mnemonic	Format	Opcode	Effect	Notes
MULF m	3/4	60	$F \leftarrow (F) * (m..m+5)$	X F
MULR r1, r2	2	98	$r2 \leftarrow (r2) * (r1)$	X
NORM	1	C8	$F \leftarrow (F)$ [normalized]	X F
OR m	3/4	44	$A \leftarrow (A) \mid (m..m+2)$	
RD m	3/4	D8	A [rightmost byte] $\leftarrow$ data from device specified by (m)	P
RMO r1, r2	2	AC	$r2 \leftarrow (r1)$	X
RSUB	3/4	4C	$PC \leftarrow (L)$	
SHIFTL r1, n	2	A4	$r1 \leftarrow (r1)$; left circular shift n bits. {In assembled instruction, $r2 = n-1$ }	X
SHIFTR r1, n	2	A8	$r1 \leftarrow (r1)$; right shift n bits, with vacated bit positions set equal to leftmost bit of (r1). {In assembled instruction, $r2 = n-1$ }	X
SIO	1	F0	Start I/O channel number (A); P X address of channel program is given by (S)	
SSK m	3/4	EC	Protection key for address m $\leftarrow (A)$ (see Section 6.2.4)	P X
STA m	3/4	0C	$m..m+2 \leftarrow (A)$	
STB m	3/4	78	$m..m+2 \leftarrow (B)$	X
STCH m	3/4	54	$m \leftarrow (A)$ [rightmost byte]	
STF m	3/4	80	$m..m+5 \leftarrow (F)$	X F
STI m	3/4	D4	Interval timer value $\leftarrow (m..m+2)$ (see Section 6.2.1)	P X
STL m	3/4	14	$m..m+2 \leftarrow (L)$	
STS m	3/4	7C	$m..m+2 \leftarrow (S)$	X
STSW m	3/4	E8	$m..m+2 \leftarrow (SW)$	P
STT m	3/4	84	$m..m+2 \leftarrow (T)$	X
STX m	3/4	10	$m..m+2 \leftarrow (X)$	
SUB m	3/4	1C	$A \leftarrow (A) - (m..m+2)$	
SUBF m	3/4	5C	$F \leftarrow (F) - (m..m+5)$	X F

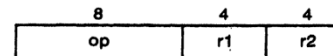
Mnemonic	Format	Opcode	Effect	Notes
SUBR r1, r2	2	94	$r2 \leftarrow (r2) - (r1)$	X
SVC n	2	B0	Generate SVC interrupt. {In assembled instruction, $r1 = n$ }	X
TD m	3/4	E0	Test device specified by (m)	P C
TIO	1	F8	Test I/O channel number (A)	P X C
TIX m	3/4	2C	$X \leftarrow (X) + 1$; (X): (m..m+2)	C
TIXR r1	2	B8	$X \leftarrow (X) + 1$; (X): (r1)	X C
WD m	3/4	DC	Device specified by (m) $\leftarrow (A)$ [rightmost byte]	P

Instruction Formats

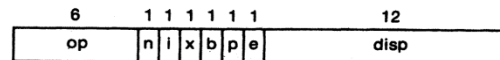
Format 1 (1 byte):



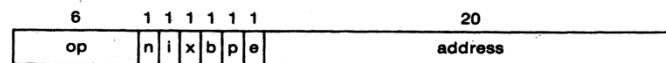
Format 2 (2 bytes):



Format 3 (3 bytes):



Format 4 (4 bytes):



Addressing Modes

The following addressing modes apply to Format 3 and 4 instructions. Combinations of addressing bits not included in this table are treated as errors by the machine. In the description of assembler language notation, *c* indicates a constant between 0 and 4095 (or a memory address known to be in this range); *m* indicates a memory address or a constant value larger than 4095. Further information can be found in Section 1.3.2.

SIC/XE 명령은 단일 모드를 사용하는 SIC 명령어와 달리 총4개의 명령어 형태를 가진다.

ASCII Character Codes

SIC/XE 명령의 플래그 비트가 주소 지정 방식을 결정한다.

Appendix B

ASCII Character Codes

The letters in the Notes column have the following meanings:

- 4 Format 4 instruction
- D Direct-addressing instruction
- A Assembler selects either program-counter relative or base-relative mode
- S Compatible with instruction format for standard SIC machine. Operand value can be between 0 and 32,767 (see Section 1.3.2 for details).

Addressing type	Flag bits n l x b p e	Assembler language notation	Calculation of target address TA	Operand	Notes
Simple	110000	op c	disp	(TA)	D
	110001	+op m	addr	(TA)	4 D
	110010	op m	(PC) + disp	(TA)	A
	110100	op m	(B) + disp	(TA)	A
	111000	op c,X	disp + (X)	(TA)	D
	111001	+op m,X	addr + (X)	(TA)	4 D
	111010	op m,X	(PC) + disp + (X)	(TA)	A
	111100	op m,X	(B) + disp + (X)	(TA)	A
	000---	op m	b/p/e/disp	(TA)	D S
	001---	op m,X	b/p/e/disp + (X)	(TA)	D S
Indirect	100000	op @c	disp	((TA))	D
	100001	+op @m	addr	((TA))	4 D
	100010	op @m	(PC) + disp	((TA))	A
	100100	op @m	(B) + disp	((TA))	A
Immediate	010000	op #c	disp	TA	D
	010001	+op #m	addr	TA	4 D
	010010	op #m	(PC) + disp	TA	A
	010100	op #m	(B) + disp	TA	A

Hex code	ASCII character	Hex Code	ASCII character	Hex code	ASCII character	Hex code	ASCII character
00	NUL	20	SP	40	@	60	`
01	SOH	21	!	41	A	61	a
02	STX	22	"	42	B	62	b
03	ETX	23	#	43	C	63	c
04	EOT	24	\$	44	D	64	d
05	ENQ	25	%	45	E	65	e
06	ACK	26	&	46	F	66	f
07	BEL	27	'	47	G	67	g
08	BS	28	(	48	H	68	h
09	HT	29	)	49	I	69	i
0A	LF	2A	*	4A	J	6A	j
0B	VT	2B	+	4B	K	6B	k
0C	FF	2C	,	4C	L	6C	l
0D	CR	2D	-	4D	M	6D	m
0E	SO	2E	.	4E	N	6E	n
0F	SI	2F	/	4F	O	6F	o
10	DLE	30	0	50	P	70	p
11	DC1	31	1	51	Q	71	q
12	DC2	32	2	52	R	72	r
13	DC3	33	3	53	S	73	s
14	DC4	34	4	54	T	74	t
15	NAK	35	5	55	U	75	u
16	SYN	36	6	56	V	76	v
17	ETB	37	7	57	W	77	w
18	CAN	38	8	58	X	78	x
19	EM	39	9	59	Y	79	y
1A	SUB	3A	:	5A	Z	7A	z
1B	ESC	3B	;	5B	[	7B	{
1C	FS	3C	<	5C	\	7C	
1D	GS	3D	=	5D	]	7D	}
1E	RS	3E	>	5E	^	7E	~
1F	US	3F	?	5F	_	7F	DEL

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